

Weekly Report: LLM + RAG Model Project

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Week Overview: 280425 - 050525

- **Project Name:** An evaluation of Retrieval-Augmented Generation (RAG) Frameworks for Disease Diagnosis
- **Project Description:** This project aims to evaluate the performance of Retrieval Augmented Generation (RAG) frameworks compared to generic LLMs for medical disease diagnosis. Specifically, we use the ICD-10 Disease Dataset, which contains a hierarchical structure of diseases and their corresponding codes to evaluate the accuracy of the RAG model in generating responses to medical queries.
- **Overview:**
 - **Goal 1:** Create Document preprocessing functions and Embedding Model for the Disease Dataset
 - **Goal 2:** Use vector database (ChromaDB) to store the embeddings
 - **Goal 3:** Retrieval of samples using similiarity search, compared with query
 - **Goal 4:** Integrate the RAG model with the LLM for generating responses
 - **Goal 5:** Test the model with sample queries
 - **Goal 6:** Compare the model's output accuracy against the Non-RAG model (LLM-only). This can be done by creating a fine-tuning model to evaluate diagnostic accuracy of the two models, trained on the hierarchical ICD-10 Disease Dataset, where partial matches at different levels of specificity have different clinical value (similar to the method in the Medfound paper).
 - **Goal 7:** Create a report on the model's performance and potential improvements

Progress

- **Completed Tasks:**
 - **Task 1:** Created data_loader.py and other tools for loading and preprocessing the dataset.
 - **Task 2:** Started on the embedding model, using the Sentence-BERT model.
 - **Task 3:** Used ChromaDB as a vector database for storing the embeddings.
 - **Task 4:** Implemented similiarity search using ChromaDB query function to retrieve top k most relevant documents based on the embedded query.

- **Task 5:** Integrated Google’s Gemma Huggingface model with varied parameter sizes (e.g., 2B, 6B) for generating responses.
- **Ongoing Tasks:**
 - **Task 1:** Create Query template for the RAG model to generate responses.
 - **Task 2:** Create the fine tuning model to evaluate the RAG model’s accuracy against the LLM-only model.
 - **Task 3:** Create a report on the model’s performance and potential improvements.